Student Performance and Study Habits Analysis

R Programming Tiny Project

Student Name	Parmar Krish
Enrollment No.	2505101200281
Division	I
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This project uses a synthetic student dataset to demonstrate an end-to-end R programming workflow for analysing study habits and academic performance.

Project Dashboard

20	71.8%	7.1 h	74.4
Students	Avg Attendance	Avg Study Time	Avg Final Score

Project Snapshot

The dashboard summarises a 20-student synthetic dataset containing attendance, study hours, assignment marks, internal marks and final examination scores.

Analysis Workflow

1	**Import & Inspect**	Read the dataset and inspect its structure.
2	**Clean**	Check missing values and prepare variables.
3	**Analyse**	Calculate descriptive statistics and correlations.
4	**Visualise**	Create charts for study habits and marks.
5	**Interpret**	Summarise patterns in student performance.

1. Introduction

This tiny project analyses a synthetic student performance dataset using R. It demonstrates data importing, preprocessing, statistical exploration, visualisation and interpretation.

2. Objectives

Import and inspect student data; check missing values; calculate descriptive statistics; study the relationship between attendance, study hours and marks; create graphs; interpret findings.

3. Dataset

The dataset contains 20 students and 7 variables: Student ID, Study Hours, Attendance Percentage, Assignment Score, Internal Marks, Final Exam Score and Result.

4. Data Preprocessing and Cleaning

Missing values were checked and incomplete records removed. Numerical variables were inspected for valid ranges and the Result field was treated as a categorical variable.

5. Statistical Analysis

Average study time is 7.1 hours per week. Average attendance is 71.8%, while the mean final examination score is 74.4. The median final score is 75.0 and the standard deviation is 12.1.

The correlation between study hours and final score is 0.76, while attendance and final score show a correlation of 0.68. These values are descriptive for this synthetic sample and do not establish causation.

6. Performance by Attendance

Students with attendance of 75% or above have a higher average final score in this sample than students below 75% attendance.

7. Study Hours Analysis

The data shows a generally positive relationship between weekly study hours and final examination scores. Students studying around 8–10 hours per week tend to have stronger scores in this sample.

Study Hours Group

Study Hours / Week	Students	Average Final Score
0–4 hours	3	57.3
5–7 hours	7	70.7
8–10 hours	7	82.9
11+ hours	3	91.3

8. Result Analysis

In the synthetic dataset, 15 students are classified as Pass and 5 as Needs Improvement. The result classification is based on the final performance values used in the dataset.

Result Summary

Result	Number of Students
Pass	15
Needs Improvement	5

9. Attendance vs Final Score

The observed correlation between attendance and final score is 0.68. This suggests a positive association in the sample, but it should not be interpreted as proof that attendance alone causes higher marks.

10. Findings

• The dataset contains 20 student records.

• Average study time is 7.1 hours per week.

• Average attendance is 71.8%.

• Average final examination score is 74.4.

• Study hours and final score have a descriptive correlation of 0.76.

• Attendance and final score have a descriptive correlation of 0.68.

• 15 students are classified as Pass and 5 as Needs Improvement.

11. Conclusion

The project demonstrates how R can import, clean, analyse and visualise student performance data. Descriptive statistics, grouped summaries and graphs help identify patterns between study habits, attendance and academic results.

12. R Functions Used

Function	Purpose
read.csv()	Import CSV data
head()	View first rows
str()	Inspect data structure
summary()	Statistical summary
is.na()	Check missing values
complete.cases()	Remove incomplete records
mean() / median()	Calculate central tendency
sd()	Calculate standard deviation
cor()	Calculate correlation
aggregate()	Create grouped summaries
table()	Frequency counts
barplot()	Create bar charts
plot()	Create scatter plots

Sample Dataset Variables

Variable	Description
Student_ID	Unique student identifier
Study_Hours	Weekly study hours
Attendance	Attendance percentage
Assignment_Score	Assignment marks
Internal_Marks	Internal assessment marks
Final_Score	Final examination score
Result	Performance category

End of Tiny Project
Prepared for R Programming